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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Name of Student:	
Date of Performance	**Date of Completion**

Experiment No. 8

Experiment Name: Design and Develop a Website

Aim/Objective

The objective of this experiment is to **design and develop a complete, multi-page website** using a combination of HTML, CSS, and PHP. You will create a functional website with a consistent layout and dynamic content, demonstrating the integration of various web technologies.

Pre-requisites

- A local web server with PHP installed (e.g., XAMPP, WAMP).
- Knowledge of HTML, CSS, and basic PHP.

Theory

A modern website is rarely a single, static HTML page. Instead, it's a collection of interconnected pages with a consistent look and feel. This is achieved by separating the core technologies:

- **HTML** provides the fundamental **structure** and content of each page.
- **CSS** provides the **presentation** and styling, ensuring a uniform layout, colors, and fonts across the entire site.
- **PHP** provides the **dynamic functionality** on the server, such as handling form submissions, interacting with databases, or generating content.

For this experiment, we will design a simple **College Portal** with three pages: a Home page, an About Us page, and a Contact page with a working form.

Problem Statement

Design and develop a website for a hypothetical college. The website should have the following structure and functionality:

1. **Home Page (index.html)**: A landing page with an introduction to the college.
2. **About Us Page (about.html)**: A page detailing the college's history and mission.
3. **Contact Page (contact.html)**: A page containing a form for users to send inquiries. This form will be handled by a PHP script.

4. **Consistent Layout**: The header, navigation bar, and footer should be consistent across all pages using a single CSS file.
5. **Form Handling**: The contact form should collect the user's name, email, and message. Upon submission, the data should be processed by a PHP script (process_form.php) that displays a confirmation message.

Steps

1. **Project Setup**: In your web server's htdocs directory, create a new folder for your website project (e.g., my_college_portal). All your files will be saved inside this folder.
2. **Create the CSS File (style.css)**: This file will define the shared styles for the entire website. Create a file named style.css and add the following code to it. This CSS creates a simple and clean layout with a fixed header, navigation, and footer.

```
body {
   font-family: Arial, sans-serif;
   margin: 0;
   padding: 0;
   background-color: #f4f4f4;
   color: #333;
}

.container {
   width: 80%;
   margin: 0 auto;
   overflow: hidden;
}

header {
   background: #333;
   color: #fff;
   padding: 1rem 0;
   text-align: center;
}

header h1 {
   margin: 0;
}

nav {
    background: #555;
   color: #fff;
    padding: 1rem 0;
}

nav ul {
   list-style: none;
   padding: 0;
   text-align: center;
```

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```
}

nav ul li {
    display: inline;
    margin: 0 15px;
}

nav a {
    color: #fff;
    text-decoration: none;
    font-weight: bold;
}

.content {
    padding: 20px;
    background: #fff;
    min-height: 400px;
}

footer {
    background: #333;
    color: #fff;
    text-align: center;
    padding: 1rem 0;
    position: relative;
    bottom: 0;
    width: 100%;
}

form {
    max-width: 500px;
    margin: 20px auto;
    padding: 20px;
    border: 1px solid #ccc;
    border-radius: 8px;
}

label, input, textarea {
    display: block;
    width: 100%;
    margin-bottom: 10px;
}

input[type="submit"] {
    background: #555;
    color: #fff;
    border: none;
    cursor: pointer;
```

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```
}
```

3. **Create the HTML Files**: Create index.html, about.html, and contact.html. Each file will have the same basic structure, linking to the style.css file.
   - index.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Home - My College Portal</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <h1>My College Portal</h1>
  </header>

  <nav>
    <ul>
      <li><a href="index.html">Home</a></li>
      <li><a href="about.html">About Us</a></li>
      <li><a href="contact.html">Contact</a></li>
    </ul>
  </nav>

  <div class="content container">
    <h2>Welcome to Our College</h2>
    <p>We are dedicated to providing a quality education and fostering a dynamic learning environment. Explore our website to learn more about our programs, faculty, and campus life.</p>
    <p>Our commitment to excellence ensures that students are well-prepared for their future careers.</p>
  </div>

  <footer>
    <p>© 2024 My College Portal</p>
  </footer>
</body>
</html>
```

   - **about.html**: (Copy the structure of index.html but change the title and content)

```
<!DOCTYPE html>
<html>
<head>
  <title>About Us - My College Portal</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <h1>My College Portal</h1>
```

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```
    </header>

    <nav>
        <ul>
            <li><a href="index.html">Home</a></li>
            <li><a href="about.html">About Us</a></li>
            <li><a href="contact.html">Contact</a></li>
        </ul>
    </nav>

    <div class="content container">
        <h2>About Our College</h2>
        <p>Founded in 1995, our college has a rich history of innovation and academic excellence. Our mission is to empower students with the knowledge and skills necessary to succeed in a rapidly changing world. We offer a wide range of degree programs and extracurricular activities.</p>
    </div>

    <footer>
        <p>© 2024 My College Portal</p>
    </footer>
</body>
</html>
```

- **contact.html**: (Copy the structure, but add the form)

```
<!DOCTYPE html>
<html>
<head>
    <title>Contact Us - My College Portal</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <header>
        <h1>My College Portal</h1>
    </header>

    <nav>
        <ul>
            <li><a href="index.html">Home</a></li>
            <li><a href="about.html">About Us</a></li>
            <li><a href="contact.html">Contact</a></li>
        </ul>
    </nav>

    <div class="content container">
        <h2>Contact Us</h2>
        <p>Have a question? Fill out the form below and we'll get back to you as soon as possible.</p>
        <form action="process_form.php" method="post">
            <label for="name">Name:</label>
```

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```
            <input type="text" id="name" name="name" required>

            <label for="email">Email:</label>
            <input type="email" id="email" name="email" required>

            <label for="message">Message:</label>
            <textarea id="message" name="message" rows="5" required></textarea>

            <input type="submit" value="Send Message">
        </form>
    </div>

    <footer>
        <p>© 2024 My College Portal</p>
    </footer>
  </body>
  </html>
```

4. **Create the PHP File (process_form.php)**: This file will handle the form data submitted from contact.html. It will then display a confirmation message.

```
<!DOCTYPE html>
<html>
<head>
  <title>Form Submission</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <h1>My College Portal</h1>
  </header>

  <nav>
    <ul>
      <li><a href="index.html">Home</a></li>
      <li><a href="about.html">About Us</a></li>
      <li><a href="contact.html">Contact</a></li>
    </ul>
  </nav>

  <div class="content container">
    <h2>Thank You!</h2>
    <p>We have received your message. Here is the information you submitted:</p>
    <ul>
      <?php
        // Check if the form was submitted
        if ($_SERVER["REQUEST_METHOD"] == "POST") {
          echo "<li><strong>Name:</strong> " . htmlspecialchars($_POST['name']) . "</li>";
          echo "<li><strong>Email:</strong> " . htmlspecialchars($_POST['email']) . "</li>";
```

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```
                echo "<li><strong>Message:</strong> " . htmlspecialchars($_POST['message']) . "</li>";
            }
        ?>
    </ul>
    <a href="contact.html">Go back to Contact Page</a>
  </div>

  <footer>
    <p>© 2024 My College Portal</p>
  </footer>
</body>
</html>
```

5. **Save and Run**: Save all five files in your my_college_portal folder. Open your browser and navigate to http://localhost/my_college_portal/index.html. Click the links to navigate between pages and test the contact form.

Output

The website should have a professional, consistent look across all three pages. The navigation links should work correctly. When you fill out the contact form and submit it, you should be redirected to a new page that displays the data you entered.

Viva Voce Questions

1. Explain the role of the <link rel="stylesheet" href="style.css"> tag. Why is it used on every page?
2. What is the purpose of the action and method attributes in the <form> tag?
3. Why is it important to use htmlspecialchars() when displaying user input from a form?
4. How would you modify this website to have five pages instead of three?
5. Describe a scenario where you would use JavaScript on this website instead of PHP.

Learning Outcome

After completing this project, you will have successfully designed and developed a complete, multi-page website. You've learned to create a consistent user experience, integrate a front-end with a back-end, and handle user input, which are core skills for any web developer.

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